

Artificial Intelligence

What is embedding?



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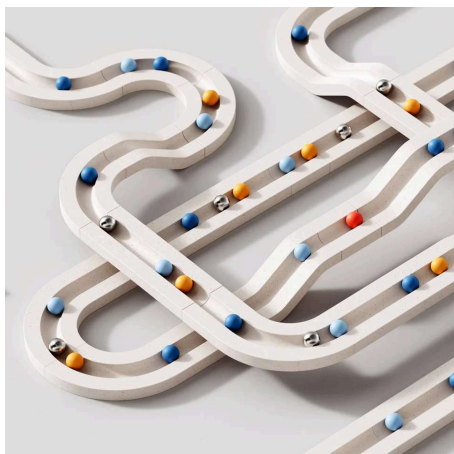
What is embedding?

Embedding is a means of representing objects as points in a continuous vector space in which the relative positions of points in space are semantically meaningful.

Embedding is a critical tool for ML engineers who build recommendation systems, chatbots, fraud detection systems. In essence, embedding enables [machine learning](#) models to understand the relationships between different objects.

Unlike other ML techniques, embeddings are learned automatically, as [neural networks](#), instead of explicitly requiring a model to learn complex patterns and relationships that are impossible for humans to identify.

For example, OpenAI's embedding implementation allows a model to understand the relationships between different words without having to understand each word in isolation. With embeddings, OpenAI can generate more accurate and contextually relevant responses to user prompts.



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How embedding wor

Most machine learning algorithms can only take numerical input. Therefore, it is necessary to convert the data into a numerical format, like creating a “bag of words” representation for text data or transforming graph data into a numerical matrix.

Objects that come into an embedding model are represented as vectors. A vector is an array of numbers (e.g. 14 numbers) where an object is along a specified dimension. The number of dimensions is a thousand or more depending on the input data. The distance between other embeddings in this n-dimensional space, and thus similarity, is determined by the length of the vector (measured by Euclidean, cosine or other).

One model, Word2Vec (word to vector), developed by Google, efficiently creates word embeddings by using a neural network. It takes a word as input and spits out an n-dimensional coordinate. We can visualize these word vectors in a three-dimensional space.

Here is how two words, “dad” and “mom” would be represented as vectors:

$$\text{“dad”} = [0.1548, 0.4848, \dots, 1.864]$$

$$\text{“mom”} = [0.8785, 0.8974, \dots, 2.794]$$

Although there is some similarity between these two words, “mom” would live in much closer proximity to “dad” in the 3D space. The cosine product (*a measure of the relative direction of two vectors in the direction they point*).

A more complex example is recommendation engines that represent users and items (e.g., movies, products, articles) as high-dimensional vectors in a latent space. These embeddings capture latent features or characteristics. The idea is to learn a representation such that the dot product of their embeddings correlates with their similarity.

Each user and item is associated with an embedding vector learned through a recommendation model during training. User and item embeddings are organized into matrices. The columns of the user matrix represent users, and the rows of the item matrix represent items.



The recommendation score for a user-item pair is calculated as the dot product of the user's embedding vector and the item's embedding vector. The higher the score, the more likely the user is to be interested in the item.

$$RecommendationScore = UserEmbedding_i \cdot ItemEmbedding_j$$

The embedding matrices are learned through a recommendation system. The model aims to minimize the difference between the predicted and actual user preferences (e.g., ratings, clicks, purchase history).

Once the model is trained, it can be used to generate recommendations for users based on their historical interactions and the predicted scores for various items.

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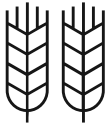
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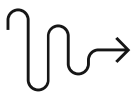
Why use embedding

Embeddings are used in various domains and a high-dimensional and categorical data into continuous space, enabling models to use meaningful patterns, relationships and semantics used in [data science](#):



Semantic representation

By mapping entities (words, images) into a continuous space, embeddings capture similarities, enabling models to understand relationships.



Dimensionality reduction

High-dimensional data, such as text or images, can be transformed into lower-dimensional representations, making it computationally efficient and easier to analyze.



Improved generalization of models

By learning meaningful representations, models can generalize better to unseen examples, making them more robust to limited labeled data.



Effective visualization

Techniques like t-SNE can be applied to visualize embeddings in two or three dimensions, revealing relationships and clusters in the data.



Efficient training in neural networks

Embedding layers are commonly used in neural networks to map categorical inputs to continuous space, facilitating backpropagation and optimization.



What objects can be

Embeddings are versatile representations that can be used for many different tasks. Here are some of the most common objects that can be embedded:

Words

Word embeddings capture the semantic relationship between words based on their usage patterns in a given language. They are represented as fixed-sized dense vector of real numbers. It is the opposite of sparse encoding, which has many zero entries.

The use of word embedding has significantly improved natural language processing (NLP) models by providing a more meaningful representation of words. These embeddings enable machines to understand the context and captures semantic nuances and contextual relationships between words. They have a wide range of applications, including sentiment analysis, text classification, and information retrieval.

Popular word embedding models include Word2Vec (Word Embedding Representation), FastText and embeddings derived from Recurrent Neural Networks (Bidirectional Encoder Representations from Transformers).

Text

Text embedding extends word embedding to represent entire documents in a continuous vector space. Text embeddings are used in various applications, such as sentiment analysis, text classification, question answering and information retrieval.

Models like Doc2Vec, USE (Universal Sentence Encoder) and Universal Sentence Encoder (Universal Sentence Encoder) have been trained on massive datasets such as Wikipedia and Google News.

Images



Image embedding is designed to capture visual content of images. Image embeddings are part of many tasks, enabling the modeling of image similarity and other visual recognition tasks.

Popular Convolutional Neural Networks (CNNs) include VGG (Visual Geometry Group), ResNet (Residual Network), and EfficientNet. These models have been pre-trained and used as powerful feature extractors.

Audio

Similar to image and text embeddings, audio embeddings are learned using various learning architectures, particularly recurrent neural networks (RNNs) or hybrid models that combine features and characteristics of audio data, allowing for similarity metrics. Audio embeddings are particularly useful for speech recognition, audio classification and music analysis.

Graphs

Graph embedding is essential for various tasks, such as node classification, link prediction, and community detection in complex networks. Graph embeddings are used in network analysis, recommendation systems, bioinformatics, and various other domains where data can be represented as a graph.

How embeddings are created

Embeddings are created through a process called embedding. The method used depends on the type of data being embedded. Here are some general steps:

1. **Choose or train an embedding model:** Select an existing model, such as GloVe, or BERT. For images, you might use pre-trained models like VGG or ResNet.



2. **Prepare your data:** Format your data in a way compatible with the embedding model. For text, this involves tokenization. For images, you may need to resize and normalize.
3. **Load or train the embedding model:** If using a pre-trained model, load the architecture. If training a new model, provide training data and labels.
4. **Generate embeddings:** For each data point, generate its corresponding embedding. For example, if using a word embedding model, you would get a vector for each word.
5. **Integrate embeddings into your application:** Use the generated embeddings in your machine learning model, or for similar tasks, depending on your specific task.

In all embedding cases, the idea is to represent data in a way that meaningful relationships are preserved. The training process involves adjusting the model parameters to minimize the difference between the predicted and actual values based on the chosen objective function. Once trained, the embeddings can be used for various downstream tasks.

Real-world examples of

Embeddings are widely used in various real-world applications. Examples include:

Natural Language Processing

- **Word embeddings in sentiment analysis:** Word embeddings are used to represent words in a continuous vector space. By analyzing the relationships between words, models can leverage these embeddings to understand the sentiment of a piece of text.
- **BERT for question answering:** BERT embeddings are used to represent the context of a question and a passage. The model can understand the context of the question and the passage to find the relevant information.



- **Text similarity with Doc2Vec:** Doc2Vec embeds similar documents. Document embeddings are used to calculate similarity between documents.

Computer vision

- **Image classification with CNNs:** CNNs, such as VGG and ResNet, are used for image classification tasks. The final layer or intermediate layers can be used to extract feature embeddings.
- **Image retrieval using CLIP:** The CLIP model from OpenAI can be used for image retrieval. This enables applications like image retrieval by text description.
- **Facial recognition with FaceNet:** FaceNet can be used for facial recognition. These embeddings are used to compare different faces.

Recommender systems

- **Collaborative filtering with embeddings:** User and item embeddings are used in collaborative filtering models. Matrix factorization is used to extract user and item embeddings to make personalized recommendations.
- **Product recommendations with word embeddings:** Product titles or reviews can be embedded using word embeddings. Products can be recommended based on the semantic similarity of their embeddings.

Cross-modal applications

- **Multimodal translation with MUSE:** MUSE (Multilingual Unsupervised Embeddings) enables cross-lingual and cross-modal understanding. It can be used for translating text between languages or connecting different modalities.
- **Cross-modal search using joint embeddings:** Joint embeddings for different modalities, such as images and text, can be used for cross-modal search. This allows one modality to retrieve results in another modality.



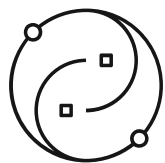
Anomaly detection

- **Network anomaly detection with graph embeddings**
can be used to detect anomalies. Changes in the behavior.
- **Fraud detection with transaction embeddings**
can identify patterns associated with fraudulent or potentially fraudulent transactions.

These examples highlight the versatility of embeddings, showcasing their ability to capture meaningful relationships across different types of data.



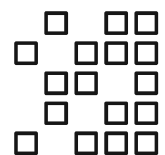
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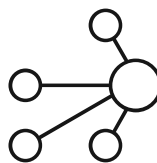
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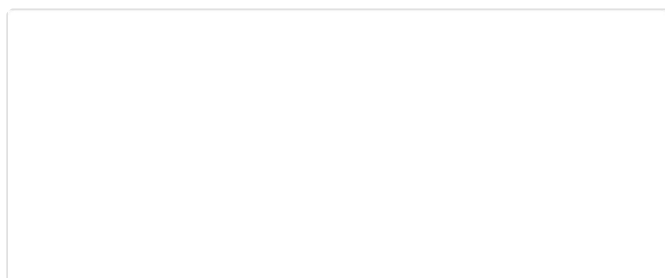
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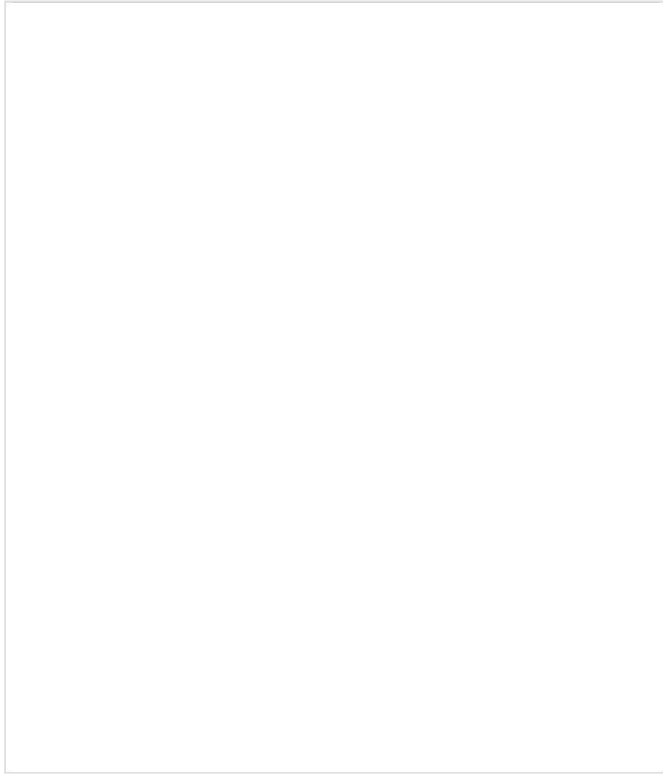
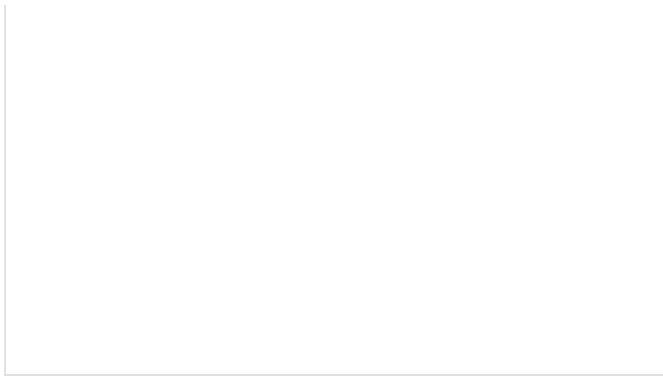
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